DATE	28-02-2026
TEAM ID	LTVIP2026TMIDS90614
PROJECT NAME	iRevolution: A Data-Driven Exploration of Apple's iPhone Impact in India Using Tableau
MAXIMUM MARKS	4 MARKS

4.3 Solution Architecture (Detailed Explanation)

The Solution Architecture of the project "iRevolution: A Data-Driven Exploration of Apple's iPhone Impact in India Using Tableau" represents the complete structural design of the system that enables end-to-end data analysis. It explains how raw data related to iPhone sales and market trends is transformed into meaningful visual insights using modern business intelligence techniques.

This architecture follows a layered analytical model, ensuring modularity, scalability, and reliability. Each layer performs a specific role and collectively supports data-driven decision-making for understanding the impact of Apple iPhone in the Indian market through Tableau.

1. Data Acquisition Layer

This layer focuses on data collection and ingestion. The project gathers data from multiple open and public sources such as:

- Kaggle datasets
- Government and telecom market reports
- Consumer surveys and industry publications

This layer ensures that the system has reliable and diverse data inputs for holistic analysis.

2. Data Integration & ETL Layer

ETL stands for Extract, Transform, Load. This layer is responsible for:

- Extracting data from different formats (CSV, Excel, JSON).
- Transforming data into a unified structure.
- Loading clean data into the analysis environment.

3. Data Modeling Layer

In this layer, data is logically structured to support analysis:

- Fact tables: sales, revenue, shipments.
- Dimension tables: year, region, iPhone model, price category.

Relationships are created between tables to enable:

- Drill-down analysis (Year → Quarter → Month).
- Cross-dimensional insights (Model vs Region vs Price).

This layer ensures data consistency and analytical flexibility.

4. Analytical Processing Layer

This layer performs the core intelligence of the system:

- Statistical aggregations (SUM, AVG, COUNT).
- Growth rate and CAGR calculations.

5. Visualization & Dashboard Layer

This is the most visible and user-centric layer:

- Interactive dashboards created using Tableau.
- Real-time filtering by year, model, and region.

Dashboards include:

- Sales trend dashboards
- Market share comparison

This layer enables quick comprehension of complex datasets.

6. User Interaction & Decision Support Layer

This layer supports human interaction:

- Users can explore data without technical knowledge.
- Decision-makers can identify:
    - High-performing regions
    - Popular models

Architectural Benefits

Aspect	Benefit
Modularity	Easy to update individual layers
Scalability	Supports large datasets
Flexibility	Works with multiple data formats
Usability	Non-technical users can analyze
Accuracy	Clean and validated data
Performance	Fast analytical queries

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